Hypercube drawings with no long plane paths

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Abstract

We study the existence of plane substructures in drawings of the d-dimensional hypercube graph Q_d. We construct drawings of Q_d which contain no plane subgraph with more than $2d-2$ edges, no plane path with more than $2d-3$ edges, and no plane matching of size more than $2d-4$. On the other hand, we prove that every rectilinear drawing of Q_d with vertices in convex position contains a plane path of length d (if d is odd) or $d-1$ (if d is even). We also prove that if a graph G is a plane subgraph of every drawing of Q_d for a sufficiently large d, then G is necessarily a forest of caterpillars. Lastly, we give a short proof of a generalization of a result by Alpert et al. [Cong. Numerantium, 2009] on the maximum rectilinear crossing number of Q_d.

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1 Introduction

Among the first graphs introduced in any undergrad graph theory course are complete graphs, complete bipartite graphs and d-dimensional hypercube graphs (which we refer to simply as d-cubes). It is therefore not surprising that drawings of complete [2, 22, 20, 27, 40, 6, 7, 8] and complete bipartite [11, 14, 1, 29] graphs are a frequent topic of research in graph drawing, with many results and interesting problems. However, one might find surprising that drawings of d-cubes have not gained such interest. As far as we are aware, except for various versions of crossing number problems [4, 23, 39, 15, 36, 16, 24, 21], drawings of d-cubes have not been studied, yet.

A d-cube Q_d is a graph with 2^d vertices labeled by binary strings of length d in which two vertices are connected by an edge if their labels differ exactly in one digit. A *rectilinear drawing* is a drawing of a graph with vertices represented by points in the plane and edges represented by line segments between the vertices. We say that a drawing is *plane* if no two edges cross. We call a drawing *convex-geometric* if it is rectilinear and its vertices are in convex position. We call a drawing *simple* if edges are represented by simple curves with no self intersections, and every two curves intersect in at most one point which is either a common vertex or a proper crossing in the interior of both edges (no tangential "touches"). (or straight-line)

We investigate the problem of finding plane subgraphs in (rectilinear) drawings of Q_d. This problem has been considered for complete and complete bipartite graphs extensively. If we are trying to find a single plane Hamiltonian path (or cycle), the problem is almost trivial for rectilinear drawings of K_n. However, if we relax the setting slightly and consider simple drawings, the problem becomes much harder. In fact, the existence of a plane Hamiltonian path in every simple drawing of K_n was conjectured in 1988 by Rafla [34], and even after almost 40 years of effort by multiple authors [2, 9, 33, 38, 17, 18, 19, 32, 25], the best known result is that every simple drawing of K_n contains a plane path of length Ω $\cancel{O}(\log(n)/\log\log(n))$ [2]. We mention that finding large plane matchings, cycles, and trees has also been considered in the literature [32, 18, 2]. In the case of rectilinear drawings, since finding one plane Hamiltonian path is easy, finding multiple such paths has been considered [26, 5]. ??

Finding plane spanning paths in rectilinear drawings of complete bipartite graphs has proven to be an interesting problem (without considering more relaxed drawings). It is

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known that there are drawings with no plane path of length more than $(4-2\sqrt{2})n+o(n)$ [13], where n is the total number of vertices. On the other hand, it is known that a plane path of length $\frac{n}{2}$ always exists [37]. Since plane Hamiltonian paths do not exist in every drawing of the complete bipartite graph, several authors have tried to give sufficient and necessary conditions for a drawing to contain a plane Hamiltonian path [12, 31, 13, 35, 28, 37].

1.1 Our results

We start by considering plane subgraphs of drawings of hypercubes in the most restrictive setting, where we only consider convex-geometric drawings. As our first result, we show that every convex-geometric drawing of Q_d contains a plane path of *length* d (that is, a path with d edges). Actually, this is an easy consequence of a famous theorem of Perles, mentioned in [30]. We include a proof of this theorem for the purpose of self-containment.

▶ **Theorem 1.** *For every $d \geq 2$, any convex-geometric drawing of Q_d contains a plane path of length at least d if d is odd, and $d-1$ if d is even.*

In Theorem 3 below, we show that Theorem 1 is tight up to a multiplicative constant. Theorem 3 follows from the following more general result (up to additive constant 1).

▶ **Theorem 2.** *For every $d \geq 3$, there is a convex-geometric drawing of Q_d which does not contain a plane subgraph with more than $2d-2$ edges.*

▶ **Theorem 3.** *For every $d \geq 4$, there is a convex-geometric drawing of Q_d which does not contain a plane path longer than $2d-3$.*

▶ **Theorem 4.** *For every $d \geq 4$, there is a convex-geometric drawing of Q_d which does not contain a plane matching on more than $2d-4$ edges.*

Note that Theorem 4 is again tight up to a multiplicative constant, since by Theorem 1, we know that every convex-geometric drawing of Q_d contains a plane path of length d or $d-1$ and, therefore, a plane matching on $\lfloor \frac{d}{2} \rfloor$ or $\lfloor \frac{d-1}{2} \rfloor$ edges, respectively.

As our last result regarding plane subgraphs of convex-geometric drawings of Q_d, we provide a necessary condition for a graph G to be a plane subgraph of every rectilinear drawing of Q_d (for sufficiently large d).

▶ **Theorem 5.** *If there is a $d \geq 2$ such that every rectilinear drawing of Q_d contains a fixed graph G as a plane subgraph, then G is a forest of caterpillars.*

We then shift our focus away from plane subgraphs of drawings of the d-cube and use a slight modification of our construction to revisit the problem of computing the *maximum rectilinear crossing number* of a d-cube. The maximum rectilinear crossing number of a graph G (denoted by $\mathrm{CR}_{\max}(G)$) is the maximum number of crossings, taken over all rectilinear drawings of G. This parameter was studied for many graph classes, such as cycles [10], complete graphs [22] and k-regular graphs [3], among others. Alpert et al. [4] investigated this question for Q_d. They proved that $\mathrm{CR}_{\max}(Q_d) \geq 2^{d-2}(2^{d-1}(d^2-2d+3)-d^2-1)$ and conjectured that this is tight. We give a significantly shorter proof of the following more general result (terms "length-regular" and "length profile" are defined in Subsection 1.2).

▶ **Theorem 6.** *Let $d \geq 2$, and let G be a d-regular graph. Then every length-regular drawing of G with length profile $(\ell_1, \ell_2, \ldots, \ell_d)$ has exactly $\frac{|V(G)|}{2} \sum_{i=1}^{d} (\ell_i - 1)\left(i - \frac{1}{2}\right)$ crossings.*

Since the drawings that we construct are length-regular, Theorem 6 can be applied ~~on~~ to them. It gives the lower bound on $\mathrm{CR}_{\max}(Q_d)$ mentioned above by a simple calculation. Interestingly, ~~even~~ although our construction is different from the one presented by Alpert et al., in the resulting drawings, the vertices lie in the same cyclic order on the convex hull ~~and they are *weakly isomorphic*, i.e., the same pairs of edges cross.~~

Lastly, we consider drawings of Q_d that are not necessarily convex-geometric and prove the following.

▶ **Proposition 7.** *Any rectilinear drawing of Q_d, for $d \geq 3$, contains a plane path of length at least 4.*

In the case of simple drawings of Q_d, we prove that Proposition 7 does not admit a straightforward generalization.

▶ **Proposition 8.** *There exists a drawing of Q_3 with no plane path of length 4.*

Our paper is organized as follows. In the next section, we give some technical definitions that we use throughout the paper. In Section 2, we derive Theorem 1 and we present our first construction, which we use to prove Theorems 2, 3, 4 and 5. In Section 3, we prove Theorem 6 and present our second construction which we use to give the above lower bound on $\mathrm{CR}_{\max}(Q_d)$. In Section 4, we prove Propositions 7 and 8. In the last section, we mention some open problems.

1.2 Preliminaries

Let $\mathcal{G}$ be a convex-geometric drawing of a graph G. The *combinatorial length*, or *length* for short, of an edge $e = uv$ in $\mathcal{G}$ is the graph distance between u and v in the cycle bounding the convex hull of the vertices of $\mathcal{G}$ (the edges of this cycle are not necessarily edges of the drawing); see Figure 1. ~~Assume that $G = Q_d$ for some d.~~ For example, for Q_d, the possible lengths of the edges are $1, 2, \ldots, 2^{d-1}$. To a vertex $v \in \mathcal{G}$, we associate a tuple $\hat{v} = (\ell_1, \ell_2, \ldots, \ell_d)$ where $\ell_1 \geq \ell_2, \geq \cdots \geq \ell_d$ are the lengths of the edges adjacent to v. If $\hat{v} = \hat{u}$ for every two vertices $u, v \in \mathcal{G}$, we say that $\mathcal{G}$ is *length-regular* and we call $\hat{v}$ the *length profile* of $\mathcal{G}$.

For a convex-geometric drawing $\mathcal{G}$ and two vertices $v, w \in \mathcal{G}$ we write $[v, w]$ for the set of vertices of $\mathcal{G}$ that are visited when traversing the convex hull of the vertices of $\mathcal{G}$ clockwise from v to w; we call this an *interval*. We write (v, w) for $[v, w] \setminus \{v, w\}$. Throughout the paper, we slightly abuse notation and identify vertices of a graph G with their representation in a drawing of G.

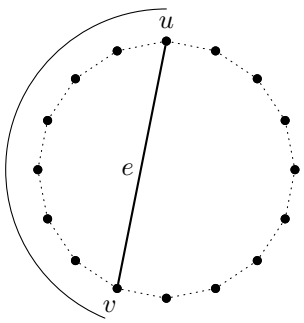

Figure 1 The length of e is 7: the number of convex hull edges within the depicted arc.

2 Plane subgraphs in convex-geometric drawings of the d-cube

We start by giving a lower bound on the length of a plane path ~~which~~ that is present in any convex-geometric drawing of the d-cube. The following result is a variation of a theorem by Perles, mentioned in [30, p. 292]. We prove it in the appendix.

▶ **Lemma 9** (⋆). *Let $k \in \mathbb{N}$, let ~~and~~ G ~~$= (E, V)$~~ be a graph, and let $n = |V(G)|$. If $|E(G)| > kn$, then ~~with $|V| = n$ and $|E| > kn$. Then,~~ any convex-geometric drawing of G contains a plane path of length at least $2k + 1$.*

▶ **Theorem 1.** *For every $d \geq 2$, any convex-geometric drawing of Q_d contains a plane path of length at least d if d is odd, and $d - 1$ if d is even.*

Proof. Q_d has $\frac{d}{2}n$ edges. If d is odd, let $k := \frac{d-1}{2}$. Then there is a plane path of length at least $2\frac{d-1}{2} + 1 = d$ by Lemma 9. If d is even, let $k := \frac{d-2}{2}$. Then there is a plane path of length $2\frac{d-2}{2} + 1 = d - 1$ by Lemma 9. ◀

We now define a particular convex-geometric drawing of Q_d, which we use to find upper bounds for the maximum sizes of various plane subgraphs in convex-geometric drawings of Q_d. In our construction, the vertices of Q_d are placed on a circle C obtained from the interval $[0, 1]$ by identifying the endpoints. With slight abuse of notation, we identify the points of C with the points in $[0, 1)$. Now, we define a drawing $\mathcal{H}_2$ of Q_2 by mapping the vertices $\overline{00} \to 0, \overline{01} \to \frac{1}{4}, \overline{11} \to \frac{1}{2}, \overline{10} \to \frac{3}{4}$. For $d \geq 3$, we construct the drawing $\mathcal{H}_d$ by taking two copies $\mathcal{H}', \mathcal{H}''$, of $\mathcal{H}_{d-1}$ and rotating $\mathcal{H}''$ so that the vertex $\overline{00\ldots0} \in \mathcal{H}''$ is placed on the point $\frac{1}{2} - \frac{1}{2^d} \in C$. We then connect every vertex in $\mathcal{H}'$ to its copy in $\mathcal{H}''$. Now, we map the vertices of Q_d to $\mathcal{H}_d$ by mapping a vertex $\overline{0x}$ to the copy of x in $\mathcal{H}'$ and $\overline{1x}$ to the copy of x in $\mathcal{H}''$. See Figure 2 for the construction of $\mathcal{H}_d$.

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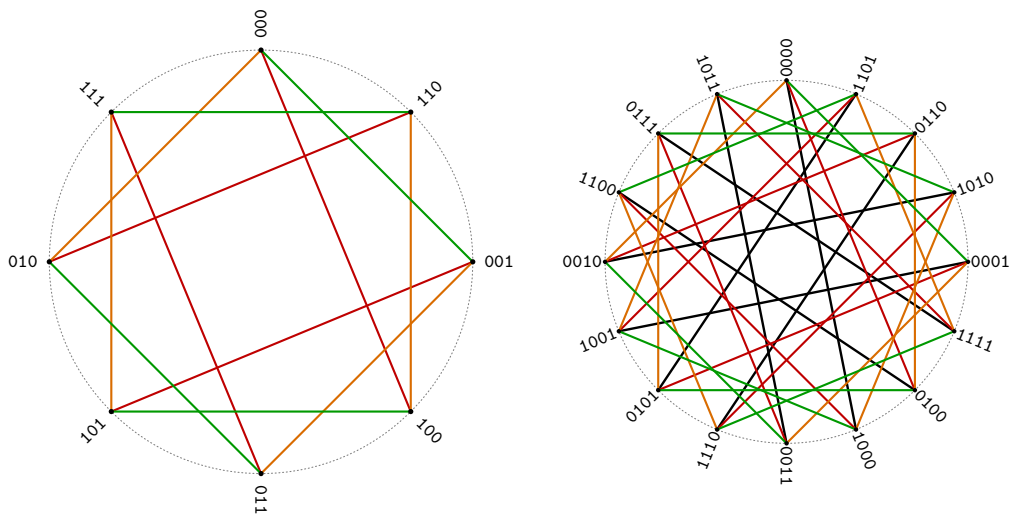

Figure 2 The construction of $\mathcal{H}_4$ (right) from $\mathcal{H}_3$ (left).

Note that, by definition, $\mathcal{H}_d$ is a length-regular drawing with length profile

$$(\ell_1, \ell_2, \ldots, \ell_d) = (2^{d-1} - 1, 2^{d-1} - 2, 2^{d-\cancel{2}\,1} - 4, \ldots, 2^{d-1} - 2^{d-2}, 2^{d-1} - 2^{d-2}).$$

For a vertex $v \in \mathcal{H}_d$, let $e_1^v, e_2^v, \ldots, e_{d-2}^v$ be the edges adjacent to v with length strictly larger than $2^{d-1} - 2^{d-2}$, sorted by length (with ~~e~~ e_1^v being the longest), and let p_v be the line through v ~~which~~ that splits C in half. We associate with each such edge e_i^v a *direction* $d(e_i^v) = \pm 1$, where

$d(e_i^v) = +1(-1)$ means that the other endpoint of e_i^v lies to the left (right) of p_v when viewed from v. We call the tuple $(d(e_1^v), d(e_2^v), \ldots, d(e_{d-2}^v))$ the *length-rotation* of v. Note that there are 2^{d-2} length-rotations that can appear in $\mathcal{H}_d$. We start by observing that all of them indeed appear in $\mathcal{H}_d$.

▶ **Lemma 10.** *For $d \geq 2$ and $\overline{v} \in \{+1, -1\}^{d-2}$, there are exactly two pairs of antipodal vertices in $\mathcal{H}_d$ with length-rotation $\overline{v}$. Thus, every possible length-rotation exists 4 times.*

Proof. We proceed by induction on d. For $d = 2$, the statement is clearly true. Assume that the statement holds for $d \geq 2$. Let $\overline{v} \in \{+1, -1\}^{d-2}$. By our inductive assumption, there are 4 vertices in $\mathcal{H}_d$ with length-rotation $\overline{v}$, denote them by u_1, u_2, u_3, u_4. Recall that $\mathcal{H}_{d+1}$ is created from two copies $\mathcal{H}, \mathcal{H}'$ of $\mathcal{H}_d$. Assume that $u_1, u_2, u_3, u_4 \in \mathcal{H}$ and let $u_1', u_2', u_3', u_4' \in \mathcal{H}'$ be their copies. Now, note that each u_i' has length-rotation $\overline{v}$ in $\mathcal{H}'$. In $\mathcal{H}_{d+1}$ we connect u_i to u_i' by an edge of length $2^{d-1} - 1$ and this edge contributes $+1$ to the length-rotation of u_i and -1 to the length-rotation of u_i'. This means that there are exactly four vertices in $\mathcal{H}_{d+1}$ with length-rotation $(\overline{v}, +1)$ and exactly four vertices with length-rotation $(\overline{v}, -1)$. Since $\overline{v} \in \{+1, -1\}^{d-2}$ was arbitrary, this finishes the proof. ◀

Lastly, we observe that the length-rotations of adjacent vertices in $\mathcal{H}_d$ agree up to the coordinate corresponding to the edge between them.

▶ **Lemma 11.** *Let $d \geq 2$ and let $e = uv$ be an edge of length $2^{d-1} - 2^m$ in $\mathcal{H}_d$ for some $m \in \{0, 1, \ldots, d-2\}$. Then if $\overline{u}$ and $\overline{v}$ are length-rotations of u and v, respectively, $\overline{u}_i = \overline{v}_i$ for all $i \in \{m+1, m+2, \ldots, d-2\}$.*

Proof. Consider the step in the process of construction of $\mathcal{H}_d$ when e is added. Then u, v are copies of the same vertex in two copies $\mathcal{H}, \mathcal{H}'$ of $\mathcal{H}_{m-1}$ and hence have the same length-rotation in $\mathcal{H}$ and $\mathcal{H}'$, respectively. The direction of any of the edges in $\mathcal{H}_m$ are not altered in any subsequent step, so the last $d - 2 - m$ coordinates of $\overline{u}$ and $\overline{v}$ remain the same. ◀

From now until the end of this section, we present several results about plane subgraphs of $\mathcal{H}_d$. We start by discussing the length of plane subpaths of $\mathcal{H}_d$.

▶ **Proposition 12.** *Let $d \geq 4$, then $\mathcal{H}_d$ contains a plane path of length $2d - 3$.*

Proof. Assume for simplicity that d is odd (otherwise the argument is similar up to some sign changes). By Lemma 10, there is a vertex $v_0 \in \mathcal{H}_d$ with length-rotation equal to $(+1, -1, +1, -1, \ldots, -1, +1)$. We begin constructing a path starting at the edge $e = v_0 v_1$ of length $2^{d-1} - 1$ adjacent to v_0. By Lemma 11, we know that the length-rotations of u_0 and u_1 agree on last $d - 3$ coordinates and hence the edges of length $2^{d-1} - 2$ adjacent to u_0 and u_1 both point to the right of u_0 and u_1 respectively, so we can add them and obtain a plane path of length 3. We can iterate this process a total of $d - 3$ times and obtain a plane path of length $2d - 5$. Let s, t be the two endpoints of this path. Recall that both s and t have two edges of length $2^{d-1} - 2^{d-2}$ adjacent to them, one of these edges has direction $+1$ and the other -1. Adding the edges with direction -1 yields a path of length $2d - 3$ as we desired. Figure 3 depicts this path in $\mathcal{H}_5$. ◀

Let us now consider an alternative construction of $\mathcal{H}_d$ which will be useful for proving our next result. Let $\mathcal{H}_2' = \mathcal{H}_2$. For $d \geq 3$, we obtain $\mathcal{H}_d'$ from $\mathcal{H}_{d-1}'$ as follows. Let $v_1, v_2, \ldots, v_{2^{d-1}}$ be vertices of $\mathcal{H}_{d-1}'$ ordered clockwise around C. Now for each vertex v_i add another vertex u_i between v_i and v_{i+1}. Now for each $i \neq j$ add an edge $u_i u_j$ if and only if

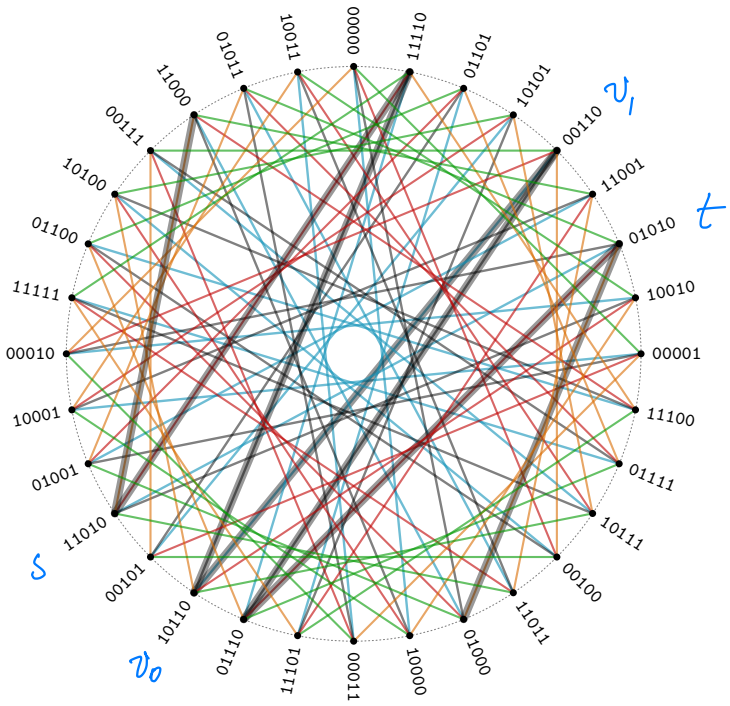

Figure 3 A plane path of length 7 in $\mathcal{H}_5$ is highlighted.

edge v_iv_j was present in $\mathcal{H}'_{d-1}$, note that if present, the two edges u_iu_j and v_iv_j necessarily cross. Lastly, for each i, we add two parallel edges $v_iu_{i+2^{d-2}}$ and $v_{i+2^{d-2}}u_i$, see Figure 4. We call the parallel edges added in the last step of construction *parallel pairs*. It is clear from the definition that if an edge e belongs to a parallel pair, then it is an edge of length $2^{d-1}-1$ that crosses every edge belonging to a parallel pair different from itself. One can observe that $\mathcal{H}'_d$ is strongly isomorphic to $\mathcal{H}_d$.

To show that Proposition 12 is in fact optimal, we prove the following more general result.

▶ **Theorem 2.** *For every $d \geq 3$, there is a convex-geometric drawing of Q_d which does not contain a plane subgraph with more than $2d-2$ edges.*

Proof. We proceed by induction on d. If $d = 3$, the statement can be checked by an easy case distinction. We assume that $d \geq 4$ and that the statement holds for $d-1$. Now we distinguish two cases based on whether G contains some edge of maximal length in $\mathcal{H}_d$ ($2^{d-1}-1$) or not. Assume that the latter is true and let $v_1, v_2, \ldots, v_{2^{d-1}}$ and $u_1, u_2, \ldots, u_{2^{d-1}}$ be as in the second definition of H_d. Recall that for any $i \neq j$ there are either zero or exactly two edges of length smaller than $2^{d-1}-1$ between (u_i, v_i) and (u_j, v_j) and if those edges exist then they necessarily cross. Therefore, G can contain at most one edge connecting a vertex in (u_i, v_i) to a vertex in (u_j, v_j) for each $i \neq j$. Lastly, note that if we replace each pair (u_i, v_i) by a single vertex w_i and we connect w_i to w_j if and only if there was an edge shorter than $\frac{1}{2} - \frac{1}{2^d}$ between (u_i, v_i) and (u_j, v_j) we obtain a drawing of Q_{d-1} which is the same as $\mathcal{H}_{d-1}$ (up to some rotation of the plane). Therefore, the subgraph of G is now a plane subgraph of $\mathcal{H}_{d-1}$ and therefore contains at most $2(d-1)-2 = 2d-4$ edges by our inductive assumption. Otherwise, note that G can contain at most 2 edges of length $2^{d-1}-1$ as every such edge belongs to a parallel pair and is crossed by all but one other edge of the same length. Removing these edges places us in the first case, and therefore G has at most

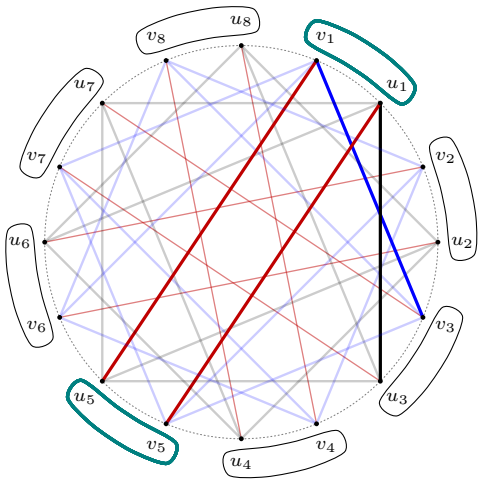

Figure 4 The alternative construction of $\mathcal{H}_d$.

$2(d-1)-2+2=2d-2$ edges. This finishes the proof. ◀

We obtain Theorem 3 as a consequence of Theorem 2.

▶ **Theorem 3.** *For every $d \geq 4$, there is a convex-geometric drawing of Q_d which does not contain a plane path longer than $2d-3$.*

Proof. A plane path in $\mathcal{H}_d$ contains at most one edge of length $2^{d-1}-1$. Then the result follows by the same argument as in Theorem 2. ◀

We now prove Theorem 4.

▶ **Theorem 4.** *For every $d \geq 4$, there is a convex-geometric drawing of Q_d which does not contain a plane matching on more than $2d-4$ edges.*

Proof. For an edge e of length ℓ in any convex drawing of a graph with n vertices, a matching cannot contain more than $\lfloor \frac{n}{l+1} \rfloor$ edges of length l. Let M be a plane matching in $\mathcal{H}_d$. The length profile in $\mathcal{H}_d$ is $\{2^{d-1}-1, 2^{d-1}-2, 2^{d-1}-4, \cdots, 2^{d-1}-2^{d-3}, 2^{d-1}-2^{d-2}, 2^{d-1}-2^{d-2}\}$. Therefore, M can contain at most 3 edges of length $2^{d-1}-2^{d-2}$ and 2 edges of any other length.

Case 1: M contains 3 edges of length $2^{d-1}-2^{d-2}$. We can show that in this case $|E(M)| \leq d-1$. First, we prove that the existence of 3 edges of length $2^{d-1}-2^{d-2}$ forbids the matching to have edges of length $2^{d-1}-1$ and $2^{d-1}-2$. Any edge $j=vw$ in the drawing has two edges of length $2^{d-1}-2^{d-2}$ on one side of j, and one on the other side of j. Then one of $[v,w]$ or $[w,v]$ contains two of the three edges of length $2^{d-1}-2^{d-2}$ and thus at least $2(2^{d-1}-2^{d-2})+3=2^{d-1}+3$ vertices of $\mathcal{H}_d$. This implies that the length of j is at most $2^{d-1}-3$ which proves the claim; see the left picture in Figure 5. Next, we show that in this case, no edge of length l for $l \in \{2^{d-1}-4, 2^{d-1}-8, \ldots, 2^{d-1}-2^{d-3}\}$ can appear more than once. The next length in our length profile that is higher than $2^{d-1}-2^{d-2}$ is $2^{d-1}-2^{d-3}$. By a similar argument to the one above, we can see that we cannot have two edges of length at least $2^{d-1}-2^{d-3}$ together with three copies of $2^{d-1}-2^{d-2}$ in the plane matching, see the middle picture in Figure 5. This proves the claim in Case 1: the size of the plane matching cannot exceed $d-1$ (3 copies of the shortest length plus one copy of each length in $\{2^{d-1}-2^{d-3}, \cdots, 2^{d-1}-2^2\}$).

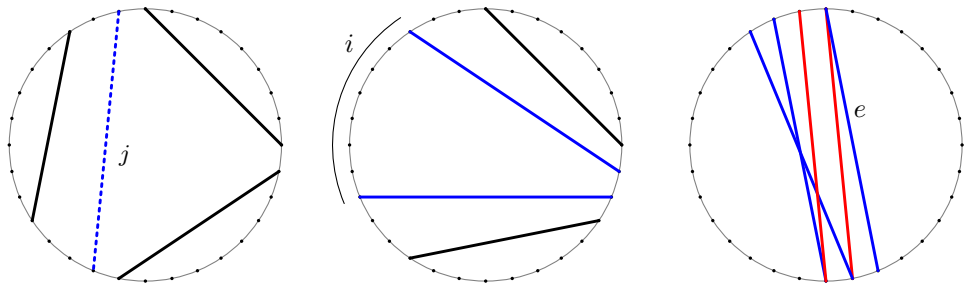

Figure 5 The cases in the proof of Theorem 4. The black edges have length $2^{d-1} - 2^{d-2}$. Left: Showing that in Case 1, an edge j in the plane matching must have length less than $2^{d-1} - 2$. Middle: Showing that in Case 1, no other edge length can appear more than once in the plane matching. The two blue lines have length at least $2^{d-1} - 2^{d-3}$ and therefore the arc i has length less than $2^d - (2(2^{d-1} - 2^{d-3}) + 1) = 2^{d-2} - 1$. Right: Proof of Case 2, showing that among the depicted lines (2 copies of the two longest lengths) only two can belong to the matching.

Case 2: M contains at most 2 edges of length $2^{d-1} - 2^{d-2}$. We claim that M contains at most two edges of length at least $2^{d-1} - 2$. We assume that there is an edge $e = vw$ of length $2^{d-1} - 2$ in M such that $[v, w]$ contains $2^{d-1} - 1$ vertices. Note that it is impossible for M to contain three edges of length $2^{d-1} - 1$, so if M contains no edges of length $2^{d-1} - 2$, we are done. If there are any other edges of length $2^{d-1} - 2$ or higher in M, both endpoints of such edges must lie in $[w, v]$. Now, there are two edges of length $2^{d-1} - 1$ and two edges of length $2^{d-1} - 2$ with both endpoints in $[w, v]$. However, one of the longer edges has either w or v as an endpoint and hence cannot be in M. The other edge of length $2^{d-1} - 1$ either shares an endpoint with or crosses the two shorter edges, so if it is contained in M, none of the shorter edges can be. Lastly, two short edges cross, and hence only one of them can be in M. See the right picture in Figure 5 for an illustration. This finishes the proof of the claim in Case 2.

By Cases 1 and 2 we conclude that the size of a maximum plane matching is at most $2d - 4$. Now, we show that there always exists a matching of this size in $\mathcal{H}_d$. ◀

We now show that the upper bound of Theorem 4 is achieved by $\mathcal{H}_d$. Therefore, to improve it, we would need a different construction. In order to prove this, we need the following claim, which is essentially just a stronger version of Lemma 10. We omit the proof because it is rather tedious and technical.

▶ Remark 13. For each vertex $x \in V(Q_d)$, its length rotation in $\mathcal{H}_d$ is given by $\overline{x} \in \{-1, +1\}^{d-2}$ with $\overline{x}_i = \begin{cases} +1, & \text{if } x_i = 0, \\ -1, & \text{if } x_i = 1, \end{cases}$ for each $i \in \{1, \dots, d-2\}$. Thus, the vertices with length rotation $\underbrace{(+1, +1, \dots, +1)}_{d-2 \text{ times}}$ are exactly the vertices $\overline{0 \dots 0x}$ with $x \in V(Q_2)$, which are the non-rotated copies of the vertices from $\mathcal{H}_2$ and thus positioned equidistantly (at distance 2^{d-2}) around the circle.

Starting at one of those four vertices and going counterclockwise around the circle (e.g. from the vertex at position 0 to the vertex at position $1 - \frac{1}{2^d}$ etc.), we see that the first $d - 2$ coordinates of the vertices are the binary encodings of the integers from $[0, 2^{d-2}]$ in increasing order. By this, we mean that if we go $0 \le m \le 2^{d-2} - 1$ steps counterclockwise from our starting vertex $\overline{\underbrace{0 \dots 0}_{d-2} x}$, we obtain a vertex y_m with $m = \sum_{i=1}^{d-2} y_i 2^{i-1}$.

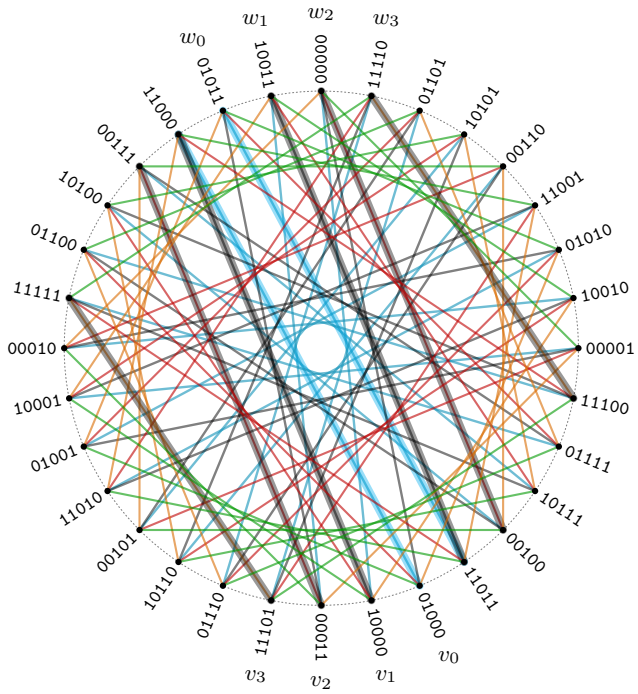

Figure 6 A plane matching of length 6 in $\mathcal{H}_5$ from the proof of Proposition 14 is highlighted in grey and vertices $v_0, \ldots, v_3$ and $w_0, \ldots, w_3$ are marked. The two edges highlighted in cyan form a parallel pair that is uncrossed by the matching and can be added to form a larger plane subgraph.

▶ **Proposition 14.** *For every $d \geq 3$, $\mathcal{H}_d$ contains a plane matching on $2d-4$ edges and a plane subgraph on $2d-2$ edges.*

Proof. Our construction is based on a claim about non-negative integers.

▷ Claim 15. For each $k \in \mathbb{N}_{\geq 1}$ there is an element $x_k \in \{0, \ldots, 2^k - 1\}$ such that for each $i \in \{0, \ldots, k-1\}$: $x_k - i \in \{0, 1, \ldots, 2^i - 1\} + 2^{i+1}\mathbb{N}_0$, i.e., the coefficient of 2^i in the binary representation of $x_k - i$ is 0.

Proof. For $k = 1$ and thus $i = 0$, we have $x_1 = 0 \in \{0\} + 2\mathbb{N}_0$. Now let us assume the claim is true for some $k \in \mathbb{N}_{\geq 1}$ and let x_k be such an element. As $x_k \in \{0, \ldots, 2^k - 1\}$ and $x_k - (k-1) \in \{0, 1, \ldots, 2^{k-1} - 1\} + 2^k\mathbb{N}_0$, we have $x_k - k \in \{-1, 0, 1, \ldots, 2^{k-1} - 2\}$. If $x_k - k \in \{0, 1, \ldots, 2^{k-1} - 2\} \subseteq \{0, 1, \ldots, 2^k - 1\} + 2^{k+1}\mathbb{N}_0$, we see that $x_{k+1} := x_k$ fulfills the claim (for the missing $i = k$ we had to check). Otherwise we have $x_k = k-1$ and we set $x_{k+1} := k-1+2^k$. For $i \in \{0, \ldots, k-1\}$, we have $x_{k+1} - i = x_k - i + 2^k \in \{0, 1, \ldots, 2^i - 1\} + 2^{i+1}\mathbb{N}_0$ and for $i = k$, we have $x_{k+1} - i = 2^k - 1 \in \{0, 1, \ldots, 2^k - 1\} + 2^{k+1}\mathbb{N}_0$. Thus, we have inductively proven the claim. ◁

We now construct our matching. Let x_{d-2} as in the Claim 15, we will identify natural numbers with their binary representation in what follows. Let v_0 be the vertex of $\mathcal{H}_d$ corresponding to $\overline{x_{d-2}00}$ and w_0 be the vertex of $\mathcal{H}_d$ corresponding to $\overline{x_{d-2}11}$, which are antipodal by Remark 13. Note that the two edges of length $2^{d-1} - 1$ adjacent to v_0 and w_0 induce a parallel pair. Now, let $v_1, v_2, \ldots, v_{d-2}$ be vertices of $\mathcal{H}_d$ consecutive on the circle C when traversing it clockwise from v_0 and define $w_1, w_2, \ldots w_{d-2}$, analogously. Now let M_v be the set of edges containing the second longest (i.e., of length $2^{d-1} - 2$) edge adjacent

to v_1, third longest (i.e., of length $2^{d-1} - 2^2$) edge adjacent to v_2 and so on. That is, for vertex v_i, we include the edge of length $2^{d-1} - 2^i$ incident to v_i in M_v. See Figure 6 for the construction of M_v. By Remark 13 and Claim 15, we know that each edge in M_v contributes $+1$ to the length rotation of the vertex v_i to which it is adjacent [1]. We show that M_v is a plane matching on $d-2$ edges. Assume that there are two edges v_iv_i' and v_jv_j' in M_v for $1 \leq i < j \leq d-2$ that are not disjoint (i.e., they either cross or share a vertex). If these edges are not disjoint, it means that v_i' lies inside the interval $(v_j, v_j']$. Note that this interval contains $2^{d-1} - 2^j$ vertices of $\mathcal{H}_d$. Furthermore, $(v_i, v_j]$ contains at most $j-i$ vertices of $\mathcal{H}_d$. Therefore, the interval $(v_i, v_j']$ can contain at most $2^{d-1} - 2^j + j - i$ vertices. On the other hand, by the definition of M_v the edge v_iv_i' is of length $2^{d-1} - 2^i$ and the interval $(v_i, v_i']$ contains exactly $2^{d-1} - 2^i$ vertices of $\mathcal{H}_d$. Hence, v_i' can lie inside the interval $(v_j, v_j']$ only if $2^{d-1} - 2^j + j - i \geq 2^{d-1} - 2^i$. This holds if and only if $2^j - 2^i \leq j - i$ which is impossible if both i, j are positive. Therefore, M_v is a plane matching on $d-2$ edges.

Analogously to M_v, we define M_w using the vertices $w_1, w_2, \ldots, w_{d-2}$. It is again a plane matching on $d-2$ edges by the same arguments as before. Furthermore, by the definition of M_v and M_w, none of the edges crosses the parallel pair induced by the edges of length $2^{d-1}-1$ adjacent to v_0 and w_0, and therefore, the disjoint union $M_v \cup M_w$ is a plane matching on $2d-4$ edges as we wanted. Additionally, $M_v \cup M_w \cup \{v_0v_0', w_0w_0'\}$ induces a plane subgraph with $2d-2$ edges. ◀

Lastly, we give a necessary condition for an abstract graph to be embeddable into every rectilinear drawing of the d-cube (for a sufficiently large d).

▶ **Theorem 5.** *If there is a $d \geq 2$ such that every rectilinear drawing of Q_d contains a fixed graph G as a plane subgraph, then G is a forest of caterpillars.*

Proof. We first show that G must be acyclic. Since Q_d is bipartite, consider a bipartition of it and let A_d, B_d be the two parts. Now, consider a drawing $\mathcal{G}$ of Q_d in which the vertices of A_d and B_d are placed on a circle in such a way that there is a line ℓ such that all vertices of A_d lie on one side of ℓ and all vertices of B_d lie on the other side. Now, if there is a plane cycle $\mathcal{C}$ of length $k \geq 4$ in $\mathcal{G}$, we can label its vertices $a_1, b_1, a_2, b_2, \ldots, a_k, b_k$ in such a way that for each i, a_ib_i, and b_ia_{i+1} are edges of $\mathcal{C}$. Consider the edges a_1b_1, b_1a_2, and b_ka_1. Since $\mathcal{C}$ is plane, all of the vertices $b_2, b_3, \ldots, b_k$ must lie in the same halfplane determined by the line through a_2 and b_1. However, a_1 must lie in the other halfplane determined by this line; otherwise, edges a_1b_1 and a_2b_2 would cross. However, this means that the edge b_ka_1 must cross the edge b_1a_2. Therefore, $\mathcal{G}$ does not contain any plane cycles. Now, to prove that G must be a forest of caterpillars, it suffices to prove that for all $d \geq 2$, $\mathcal{H}_d$ does not contain the graph G_0, obtained by subdividing each edge of $K_{1,3}$ once, as a plane subgraph. Assume for contradiction that there is some d such that $\mathcal{H}_d$ contains a plane copy of G_0. Let v be the unique degree three vertex in G_0, and let $e_1 = vw_1, e_2 = vw_2, e_3 = vw_3$ be the three edges adjacent to v in G_0 and let w_1', w_2' and w_3' be the leaves in G_0 adjacent to w_1, w_2 and w_3, respectively. Without loss of generality, we can assume that in $\mathcal{H}_d$, G_0 is embedded so that e_1, e_2, e_3 appear in this order clockwise around v. Now, up to symmetry, the remaining edges of G_0 are embedded in one of the three ways depicted in Figure 7. For the remainder of the proof, we say that a vertex w_i' lies to the left of w_i if it lies in the interval (w_i, v), and that it lies to the right of w_i if it lies in the interval (v, w_i).

[1] This is not technically correct for v_{d-2} as it is adjacent to an edge of length $2^{d-1} - 2^{d-2}$ which is not included in the length-rotation but as there are two such edges, we can assume that we are using the one that would contribute $+1$ (i.e., the one that is to the left of v_{d-2})

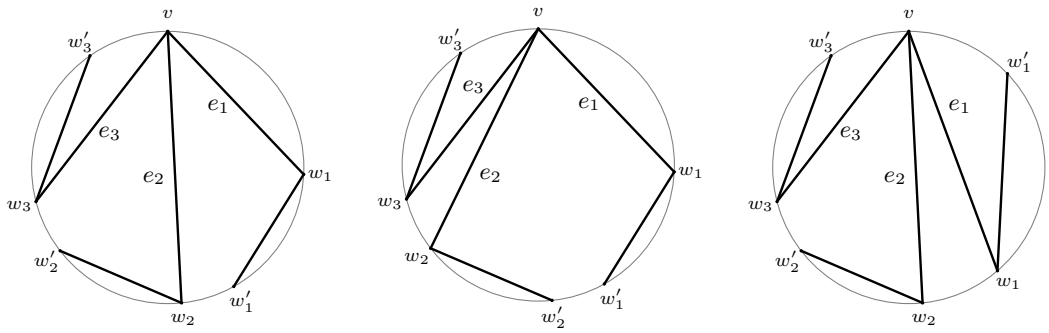

Figure 7 Three possible cases of G_0 in the proof of Theorem 5

Case 1: w_i' is placed to the left of w_i for every $i \in \{1, 2, 3\}$, as in the left part of Figure 7. In this case, the edges e_1 and e_3 are of length at least $2^{d-1} - 2^{d-2}$, and hence the vertices $w_1, w_1', w_2, w_2', w_3$ are all contained in the interval $[w_1, w_3]$, which contains at most

$$2^d - (2^{d-1} - 2^{d-2}) - (2^{d-1} - 2^{d-2}) + 1 = 2^{d-1} + 1$$

vertices of $\mathcal{H}_d$. However, the edges w_1w_1' and w_2w_2' are both of length at least $2^{d-1} - 2^{d-2}$, and therefore the interval $[w_1, w_2']$ needs to contain at least

$$2(2^{d-1} - 2^{d-2}) + 1 = 2^{d-1} + 1$$

vertices of $\mathcal{H}_d$, which is impossible since $[w_1, w_2']$ is properly contained in the interval $[w_1, w_3]$.

Case 2: the leaves w_1', w_3' are to the left of w_1 and w_3, while w_2' is placed to the right of w_2, as in the middle part of Figure 7. Then, the vertices w_1, w_1', w_2', w_2 are all contained inside the interval $[w_1, w_2]$. As before, the edge e_1 is of length at least $2^{d-1} - 2^{d-2}$ and e_2 is of length at least $2^{d-1} - 2^{d-3}$ so the interval $[w_1, w_2]$ contains at most $2^{d-2} + 2^{d-3} + 1$ vertices of $\mathcal{H}_d$. On the other hand, the edges w_1w_1' and w_2w_2' are both of length at least $2^{d-1} - 2^{d-2}$ and $w_1' \neq w_2'$ and therefore, for the edges w_1w_1' and w_2w_2' to be embedded without crossing, it would be necessary for the interval $[w_1, w_2]$ to contain at least $2^{d-1} + 1$ vertices of $\mathcal{H}_d$, contradicting our previous calculations.

Case 3: w_2', w_3' are placed to the left of w_2 and w_3, while w_1' is placed to the right of w_1, as in the right part of Figure 7. Edges w_1w_1' and w_3w_3' are both of length at least $2^{d-1} - 2^{d-2}$, and hence, edges e_1 and e_3 are both of length at least $2^{d-1} - 2^{d-3}$. Vertices w_1, w_2, w_2', w_3 all lie inside the interval $[w_1, w_3]$, which contains at most $2^{d-2} + 1$ vertices of $\mathcal{H}_d$. Since the edge w_2w_2' is of length at least $2^{d-1} - 2^{d-2} = 2^{d-2}$ and the vertices w_2, w_2' lie in the interval (w_1, w_3), it is necessary that this interval also contains at least $2^{d-2} + 1$ vertices of $\mathcal{H}_d$, which is impossible since $(w_1, w_3) \subsetneq [w_1, w_3]$.

By the previous discussion, if G is a graph that can be embedded in every drawing of Q_d for sufficiently large d, then G must be acyclic and cannot contain G_0 as a subgraph, and therefore G must be a forest of caterpillars. ◀

3 Maximum Rectilinear Crossing number of the d-cube

similarly as

We now define another drawing of Q_d, (in a similar way to the way) we defined $\mathcal{H}_d$. Let C be as before and let $\mathcal{R}_2$ be a drawing of Q_2 defined by mapping the vertices of Q_2 as follows: $\overline{00} \to 0, \overline{01} \to \frac{1}{2}, \overline{11} \to \frac{1}{4}, \overline{10} \to \frac{3}{4}$. For $d \geq 3$, we obtain $\mathcal{R}_d$ from $\mathcal{R}_{d-1}$ in the same way we obtained $\mathcal{H}_d$ from $\mathcal{H}_{d-1}$; see Figure 8. As mentioned before, Alpert et al. [4] constructed a

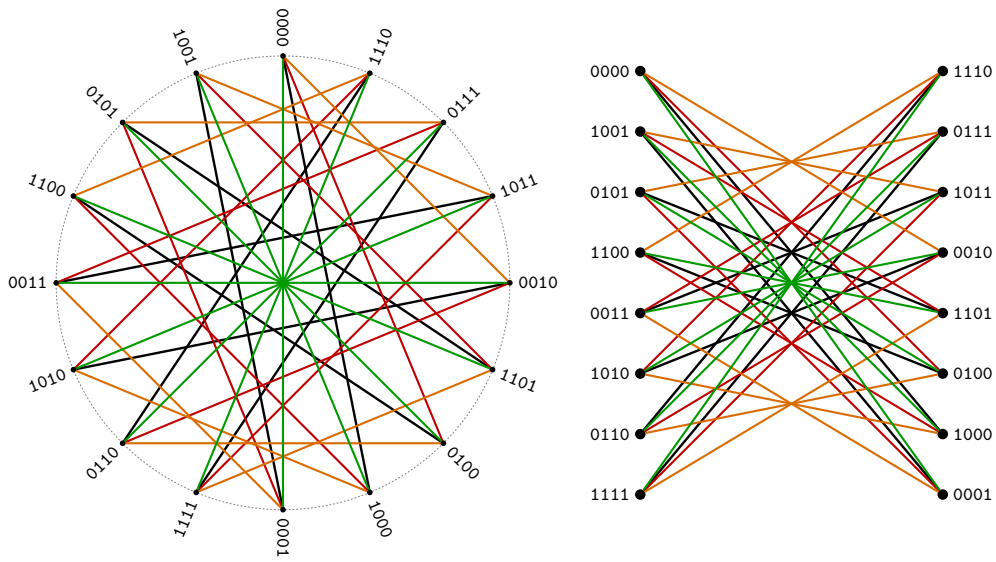

Figure 8 Left: $\mathcal{R}_4$; Right: The drawing from the construction of Alpert et al. for Q_4.

drawing strongly isomorphic to $\mathcal{R}_d$ and conjectured that this drawing achieves $CR_{\max}(Q_d)$. We prove the following more general result. Than the conjecture??

▶ **Theorem 6.** *Let $d \geq 2$, ~~and let~~ if G ~~be~~ is a d-regular graph, then every length-regular drawing of G with length profile $(\ell_1, \ell_2, \ldots, \ell_d)$ has exactly $\frac{|V(G)|}{2} \sum_{i=1}^{d} (\ell_i - 1)\left(i - \frac{1}{2}\right)$ crossings.*

Proof. Let, $d \geq 2$ and let $\mathcal{D}$ be a length-regular drawing of G with length profile $(\ell_1, \ell_2, \ldots, \ell_d)$. Note that if $\ell_i = \ell_{i+1}$ for some i (i.e., if the drawing contains all possible edges of this length), we consider half of the edges of this length as edges of length ℓ_i and the other half as edges of length ℓ_{i+1} in a way that each vertex is incident to one of each of them. An edge $e = vw$ of length ℓ_i and an edge $f = xy$ of length $\geq \ell_i$, cross precisely if one of x, y lies in the interval (v, w) and the other lies in the interval (w, v). As the drawing is length-regular, for each of the $\ell_i - 1$ vertices between v and w there is an edge for each length, in particular for each ℓ_j with $j < i$. Thus, we count in total $(\ell_i - 1)(i - 1)$ crossings of e with edges of length $\ell_1, \ldots, \ell_{i-1}$ and $(\ell_i - 1)$ crossings of e with edges of length ℓ_i. Adding the crossings of all $\frac{|V(G)|}{2}$ edges of length ℓ_i with edges of length $\geq \ell_i$, we obtain $\frac{|V(G)|}{2}(\ell_i - 1)(i - \frac{1}{2})$ crossings, where the $i - \frac{1}{2}$ appears since crossings between two edges of length ℓ_i are counted twice. Summing over all lengths, we see that the total number of crossings in $\mathcal{D}$ is $\frac{|V(G)|}{2} \sum_{i=1}^{d} (\ell_i - 1)(i - \frac{1}{2})$. ◀

▶ **Corollary 16.** *As the drawing $\mathcal{R}_d$ is length-regular with lengths $\ell_1 = 2^{d-1}$ and $\ell_i = 2^{d-1} - 2^{i-2}$ for $2 \leq i \leq d$, ~~the maximum rectilinear crossing number~~ $CR_{\max}(Q_d)$ ~~is at least~~ $\geq$ $2^{d-2}(2^{d-1}(d^2 - 2d + 3) - d^2 - 1)$.*

We note that Theorem 6 allows us to recover simpler proofs of other results about maximum rectilinear crossing numbers. For example, *star-like* drawings of d-regular graphs presented in [3] are also length-regular. These drawings are known to maximize the maximum rectilinear crossing number taken over all d-regular graphs with a given number of vertices.

4 Plane paths in rectilinear and simple drawings of the d-cube

▶ **Proposition 7.** *Any rectilinear drawing of Q_d, for $d \geq 3$, contains a plane path of length at least 4.*

Proof. Consider an arbitrary rectilinear drawing $\mathcal{D}$ of Q_3. We consider two cases.

Case 1: $\mathcal{D}$ contains a plane 4-cycle. Let the vertices of this plane 4-cycle be a, b, c, d, we need to consider two subcases, depending on whether a, b, c, d form a convex or non-convex 4-gon.

Subcase 1.1: a, b, c, d form a convex 4-gon. Let a', b', c', d' be the remaining vertices of Q_3 so that xx' is an edge for every $x \in \{a, b, c, d\}$. For each vertex $x \in \{a, b, c, d\}$ there are exactly two plane paths p_x, q_x in the 4-gon $\{a, b, c, d\}$ that have x as an endpoint. For example $p_a = d, c, b, a$ and $q_a = b, c, d, a$. Now consider the paths $p'_a = d, c, b, a, a'$ and $q'_a = b, c, d, a, a'$. If any of these paths is plane, we are done, so we can assume that the edge aa' crosses one of the edges bc or cd. Therefore, a' must lie in the section of the plane defined by the angle $\angle dab$ and outside of the convex hull of $\{a, b, c, d\}$, see Figure 9, we call this a *plausible region* for a'. Similarly, we can conclude the plausible regions for b', c', d'. Now, if the edge $a'b'$ does not cross the edge cb, we have a plane path of length 4 given by b', a', a, b, c. Otherwise, if $b'a'$ crosses bc, note that it is impossible for $a'd'$ to cross dc and therefore d', a', a, d, c is a plane path of length 4 in $\mathcal{D}$.

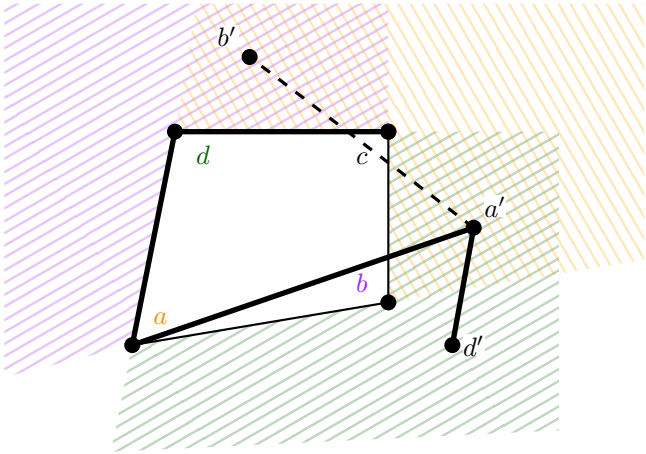

Figure 9 Subcase 1.1 in the proof of Proposition 7. The regions in which vertices a', c', d' can be placed are shaded, plane path d', a', a, d, c is drawn in bold.

Subcase 1.2: a, b, c, d form a non-convex 4-gon. Assume that a, b, c, d are placed as in Figure 10, i.e., that c is inside the triangle $\triangle abd$. Let a', b', c', d' and p_x, q_x be defined as in the previous subcase. By the same arguments as before, we can find plausible regions in which a', b', d' can lie, which can be seen in Figure 10. Again, if the edge $b'a'$ does not cross the edge bc, the path b', a', a, b, c is plane and we are done. Otherwise, it is impossible for $a'b'$ to cross bc and $a'd'$ to cross dc so the path d', a', a, d, c is a plane path of length 4 in $\mathcal{D}$.

Case 2: There is no plane 4-cycle in D. Let a, b, c, d be an arbitrary 4-cycle in D and assume that the edges ab and cd cross. Let a', b', c', d' be defined as in the previous cases. Additionally, since a, b, c, d is not plane, for each $x \in \{a, b, c, d\}$ there is a unique plane path of length 3 with x as an endpoint in the 4-cycle a, b, c, d, denote this path p_x. Using similar arguments as in previous cases, we can determine plausible regions for a', b', c', d', which can be seen in Figure 11. Furthermore, if the edge $c'd'$ does not cross both of the

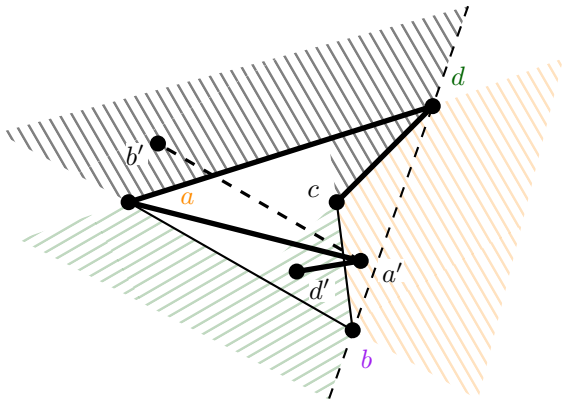

Figure 10 Subcase 1.2 in the proof of Proposition 7.

edges ad, bc, then either a, d, d', c', c or d, d', c', c, b is a plane path of length 4, so we assume that $c'd'$ crosses both of these edges, similarly, we assume that $a'b'$ crosses both of these edges. Consider the 4-cycle c', d', d, c. We know that this 4-cycle is not plane, so either $c'd'$ crosses cd or $c'c$ crosses $d'd$.

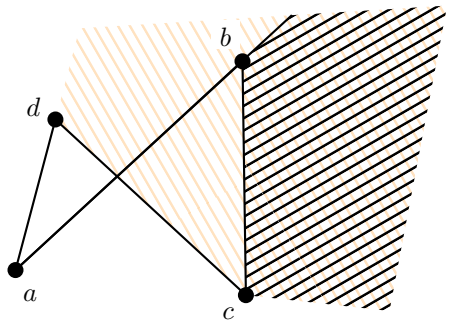

Figure 11 Setup of case 2 in the proof of Proposition 7. Before considering the fact that $a'b'$ must cross both edges ad and bc, we can conclude that a' can be placed into the orange region. With additional information, a' can only be placed in the dark red shaded region.

Subcase 2.1: $c'c$ crosses $d'd$. In this case, $c'd'$ does not cross either ad or bc, which contradicts our assumptions.
Subcase 2.2: $c'd'$ crosses cd. Now, a and b lie in different halfplanes determined by line ℓ through c', d'. If a' and b' are both in the same halfplane determined by ℓ, then the 4-cycle a', b', c', d' is plane, contradicting the assumption of this case. So, we can assume that a' lies in the same halfplane determined by ℓ as b and b' lies in the same halfplane determined by ℓ as a. Recall that we assume that $a'b'$ crosses both ad and bc. Therefore, we are in the situation as in Figure 12, so the path c', b', b, a, a' is a plane path of length 4. ◀

Finally, we construct a simple drawing of Q_3 with no plane path of length 4. Our construction can be seen in Figure 13 below.

▶ **Proposition 8.** *There exists a drawing of Q_3 with no plane path of length 4.*

Proof. Let $\mathcal{D}$ be the drawing in Figure 13, we refer to edges by their color (red/blue/black) and to vertices by their label in the figure. We will show that $\mathcal{D}$ does not contain a plane path of length 4. Suppose for contradiction that there is a plane path p of length 4 in $\mathcal{D}$.

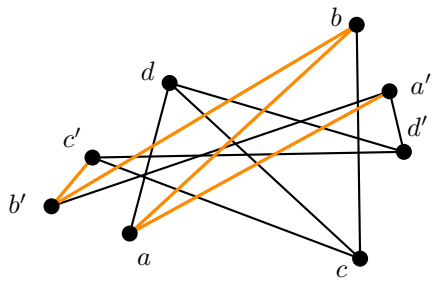

Figure 12 Subcase 2.2 in the proof of Proposition 7, the plane path of length 4 $\langle c', b', b, a, a' \rangle$ is drawn in orange.

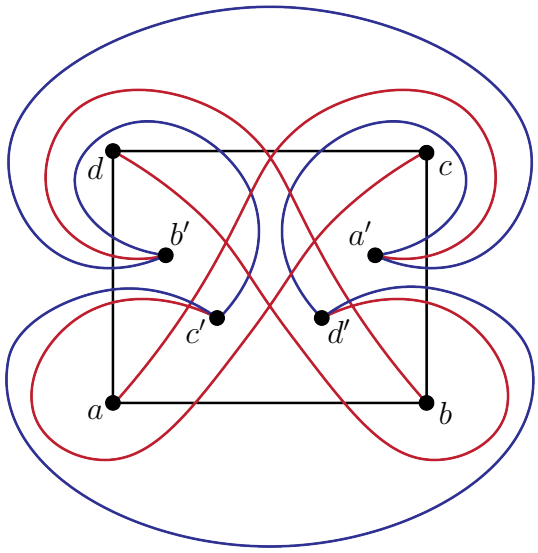

Figure 13 Simple drawing of Q_3 with no plane path of length 4. It's unusual that a cycle of length 4 is not considered a path of length 4.

Observe that every red edge in $\mathcal{D}$ crosses every other red edge and therefore p can contain only one red edge. We now differentiate between 4 cases based on the order of edges within p.

Case 1: p consists of three consecutive blue edges and one red edge. Observe that each red edge crosses the blue cycle at least once. For example, the edge bb' crosses the edge $a'd'$. Now, if p contains the edge bb', by the assumptions of the case we are in, we can see that $p = a, d', c', b', b$ and is hence not plane. Similar analysis holds for the other three red edges.

Case 2: p consists of two consecutive blue edges, one red edge and one black edge. Assume that aa' is the single red edge in p. Since it is preceded by two blue edges, we know that $p = c', d', a', a, x$ where $x \in \{b, d\}$. However, both the edges ab and ad cross one of the edges $c'd'$ or $d'a'$, hence p is not plane. Similar analysis holds for the other red edges.

Case 3: p consists of one blue edge, one red edge and two consecutive black edges. Note that every red edge crosses exactly two black edges, so the only black edges not crossed by the red edges share a vertex with it. Therefore, it is impossible for a red edge to be preceded by more than one black edge in any plane path.

Case 4: p consists of one red edge and three consecutive black edges. Same as Case 3. ◀

5 Discussion and Open Problems

To the best of our knowledge, we were the first to investigate existence of plane substructures in drawings of Q_d. While we ~~gave~~ have given partial or full answer to some questions, many ~~still~~ remain open. We highlight the ones that we consider the most interesting here.

Firstly, our study was mostly concerned with convex-geometric drawings of Q_d, and we did not investigate general rectilinear or simple drawings of Q_d in much detail. Investigating the problems in these settings may pose some interesting challenges. We attempted to find a counterpart to Theorem 1 for general rectilinear drawings. However, the bound that we obtain in Proposition 7 is considerably worse than that obtained from Theorem 1 for convex-geometric drawings. Note that the upper bound on the length of the longest plane path from Theorem 3 is still valid for general rectilinear drawings, so the gap between the lower and upper bounds is very large. It would be interesting to improve any of these bounds.

For simple drawings, we know even less. We constructed a drawing of Q_3 that does not contain a plane path of length more than 3. Note that a path of length 3 can always be found already in Q_2. It is possible (but it would be surprising) that our construction can be extended to higher dimensions. This would be particularly interesting since determining the maximum number of edges that a simple drawing can have without containing a plane path of given length was investigated recently by Keszegh et al. [25], so such constructions would be interesting from this perspective as well.

Another intriguing problem would be to close the gap between the bounds of Theorems 1 and 2. Currently, we know that every convex-geometric drawing of Q_d contains a plane path of length at least $d-1$ and that some drawings ($\mathcal{H}_d$) do not contain plane paths of length larger than $2d-3$. We are not sure which (if any) of these bounds is tight.

Lastly, we mention the conjecture of Alpert et al. [4], which says that the maximum crossing number $\mathrm{CR}_{\max}$ of Q_d is obtained by the drawing $\mathcal{R}_d$. Since this conjecture seems to be hard to solve, we think that verifying it for the more restricted classes of convex-geometric or even length-regular convex-geometric drawings of Q_d might be more approachable. A more ambitious direction might be to consider the maximum crossing number of Q_d for general simple drawings.

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A Proof of Lemma 9

▶ **Lemma 9** ($\star$). *Let $k \in \mathbb{N}$ and $G = (E, V)$ be a graph with $|V| = n$ and $|E| > kn$. Then, any convex-geometric drawing of G contains a plane path of length at least $2k + 1$.*

Proof. Let Γ be a convex-geometric drawing of G. We construct a sequence of convex-geometric drawings $\Gamma_0, \Gamma_1, \Gamma_2, \ldots, \Gamma_k$ of subgraphs of G in the following way. We start with $\Gamma_0 = \Gamma$. Then, we obtain Γ_1 from Γ_0 by removing the leftmost edge adjacent to each vertex. Then, for $i \in \{2, \ldots, \lfloor \frac{k-1}{2} \rfloor\}$ we obtain Γ_i alternatingly removing the rightmost and leftmost edge at every vertex. As every step deletes at most n edges, it is clear by our assumptions that Γ_k contains at least one edge $e = uv$. In the following we assume that k is odd and therefore Γ_k was obtained from Γ_{k-1} by removing the leftmost edge at every vertex. But since e was not removed from Γ_{k-1}, it was not the leftmost edge of any of its endpoints. Let $l_u = uu'$ and $l_v = vv'$ be the leftmost edges of u and v respectively in Γ_{k-1}. Then, l_u, e, and l_v together form a plane path with three edges. Now, since l_u and l_v were not removed in the step from Γ_{k-2} to Γ_{k-1}, it means that l_u and l_v were not the rightmost edges of u' and v' and we can again find two edges in Γ_{k-2} that extend this path. Since we can repeat this argument k times, in the end we obtain a path consisting of $2k + 1$ edges. ◀